

Consider the planar motion of velocity components $v_x = ax$ and $v_y = by$, where a and b are known constants.

1. If the flow is irrotational compute the associated velocity potential.
2. Compute the value of b for which the flow corresponds to that of an incompressible fluid (use that value in the solution of the following questions).
3. Obtain the stream lines and represent the result schematically.
4. Determine the trajectory and the path of a fluid particle initially located at $x_0 = 0$ and $y_0 = 1$. How long does it take to reach the origin?
5. Calculate the acceleration of fluid particles located along the axis $y = 0$.
6. Find the stream function for the flow.
7. Compute the rate-of-strain tensor.
8. Investigate the evolution of a fluid element whose initial shape is that of a square of sides dl ,

$$1) \bar{\omega} = \nabla \wedge \bar{v} = \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \bar{e}_z = 0, \quad \left. \begin{array}{l} v_x = \frac{\partial \psi}{\partial y} = ax \\ v_y = \frac{\partial \psi}{\partial x} = by \end{array} \right\} \quad \psi = a \frac{x^2}{2} + b \frac{y^2}{2} + \psi_0$$

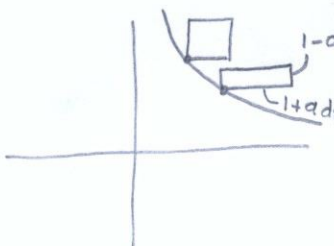
$$2) \nabla \cdot \bar{v} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = a + b = 0 \Rightarrow b = -a \quad \left| \quad 3) \frac{dx}{v_x} = \frac{dy}{v_y} \Rightarrow \frac{dx}{x} = -\frac{dy}{y} \quad xy = x_0 y_0 \right.$$

$$4) \begin{array}{l} \frac{dx}{dt} = ax \\ \frac{dy}{dt} = -ay \end{array} \quad \begin{array}{l} t=0: \\ x=0 \\ y=1 \end{array} \quad \begin{array}{l} x = C_x e^{at} \\ y = C_y e^{-at} \end{array} \Rightarrow \begin{array}{l} x=0 \\ y=e^{-at} \end{array} \quad y \rightarrow 0 \text{ when } t \rightarrow \infty$$

$$5) \frac{D\bar{v}}{Dt} = \frac{\partial \bar{v}}{\partial t} + \bar{v} \cdot \nabla \bar{v} = \left(v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} \right) \bar{e}_x + \left(v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} \right) \bar{e}_y = a^2 x \bar{e}_x + a^2 y \bar{e}_y \xrightarrow{y=0} \frac{D\bar{v}}{Dt} = a^2 x \bar{e}_x$$

$$6) \begin{array}{l} v_x = ax = \frac{\partial \psi}{\partial y} \\ v_y = -ay = -\frac{\partial \psi}{\partial x} \end{array} \quad \boxed{\psi = axy + \psi_0} \quad \xrightarrow{\text{STREAMLINES}} \psi = \text{constant} \Rightarrow xy = \text{constant}$$

$$7) \bar{T}_d = \frac{1}{2} (\nabla \bar{v} + \nabla \bar{v}^T) = \begin{bmatrix} a & 0 \\ 0 & -a \end{bmatrix} = \nabla \bar{v} \quad \left(\bar{T}_r = 0 \text{ BECAUSE } \bar{\omega} = 0 \right)$$

8)  $\begin{array}{l} \text{NO ROTATION, NO SHEAR STRAIN} \\ dl(1,0) \Rightarrow dl(1,0) + dl(1,0) \cdot (\nabla \bar{v}) dt = dl(1+adt, 0) \\ dl(0,1) \Rightarrow dl(0,1) + dl(0,1) \cdot (\nabla \bar{v}) dt = dl(0, 1-adt) \end{array}$